

Retinal Disease Classification: CNN vs. ResNet50 Evaluation

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1 Dataset and Class Distribution

This report details the retinal disease classification project, aiming to classify fundus images into eight distinct categories. Figure 1 displays sample images from each of the eight disease classes.

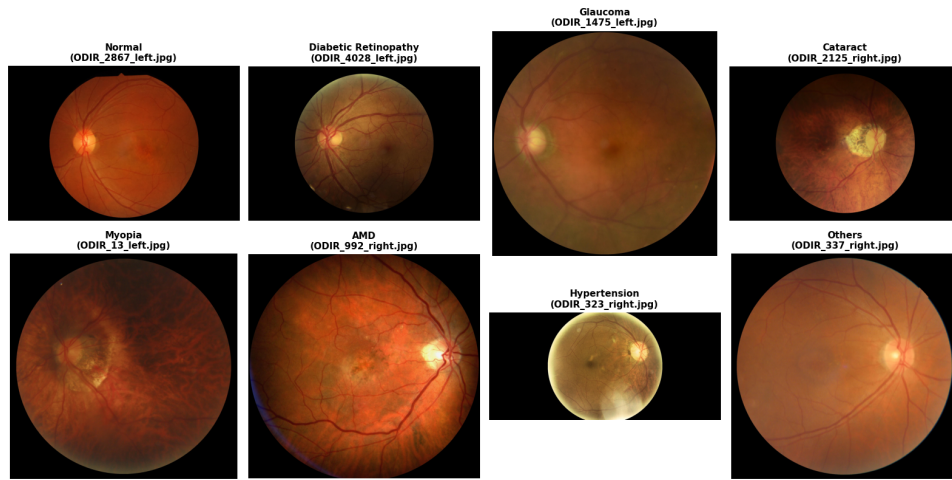


Figure 1: Sample fundus images for the eight retinal disease categories.

An analysis of the dataset revealed a notable class imbalance. The exact sample counts per class are detailed in the class distribution plot shown in Figure 2. To address this imbalance, class weights were applied to the loss function during model training.

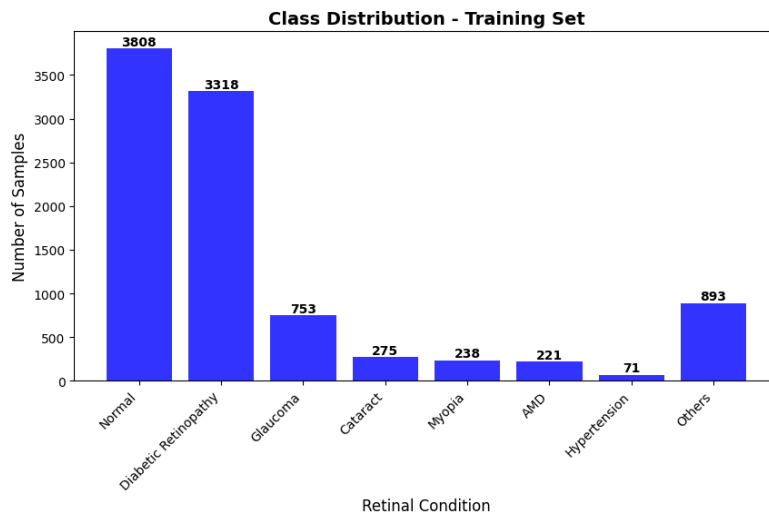


Figure 2: Distribution of samples across the eight disease classes.

2 Custom CNN Training and Evaluation

A Convolutional Neural Network (CNN) was constructed and trained from scratch. Figure 3 shows the training and validation metrics. The plot indicates that the training loss decreases over the epochs, demonstrating reasonable training convergence.

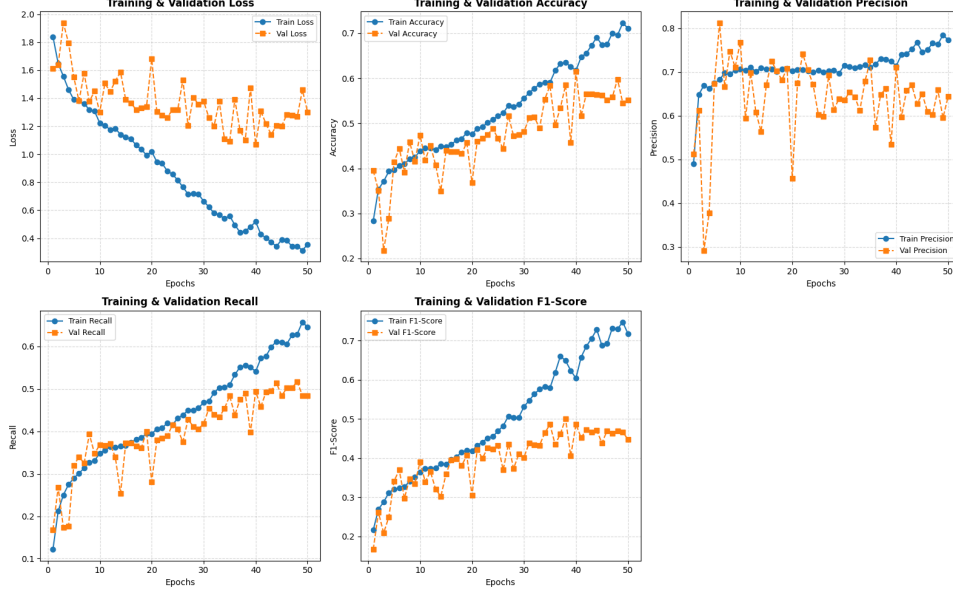


Figure 3: Training and validation loss and accuracy over epochs for the custom CNN.

The model was evaluated on a test set of 1,065 images, achieving an overall accuracy of 0.53. The per-class performance metrics are recorded in Table 1.

Table 1: Per-Class Classification Report (CNN from Scratch)

Class	Precision	Recall	F1-Score	Support
Normal	0.78	0.54	0.63	423
Diabetic Retinopathy	0.78	0.53	0.63	369
Glaucoma	0.45	0.69	0.55	84
Cataract	0.42	0.81	0.56	31
Myopia	0.32	0.77	0.45	26
AMD	0.07	0.36	0.12	25
Hypertension	0.25	0.12	0.17	8
Others	0.19	0.28	0.23	99
Accuracy				0.53
Macro Avg	0.41	0.51	0.42	1065
Weighted Avg	0.66	0.53	0.57	1065

3 Fine-Tuned ResNet50 Training and Evaluation

A pretrained ResNet50 model was fine-tuned on the same dataset. During the model configuration, it was determined that using the specific ResNet preprocess function (`tf.keras.applications.resnet50`) is critical. Prior to applying this function, the model failed to learn effectively and performed worse than the custom CNN, as shown by the stagnant validation metrics in Figure 4.

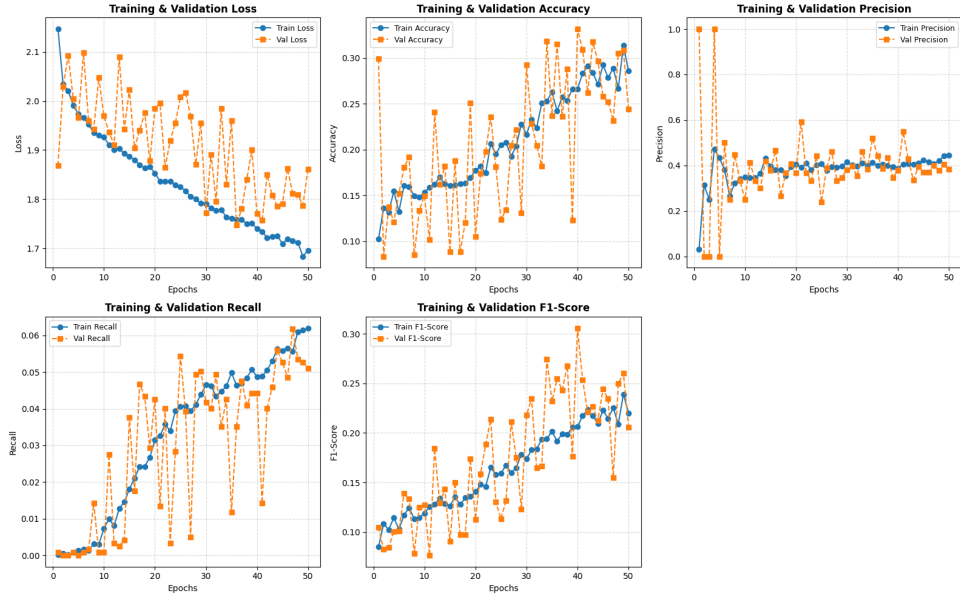


Figure 4: Training metrics displaying poor learning due to incorrect image preprocessing.

With the correct preprocessing implemented, the model successfully completed its training. The final training and validation loss and accuracy metrics over the full epochs are shown in Figure 5.

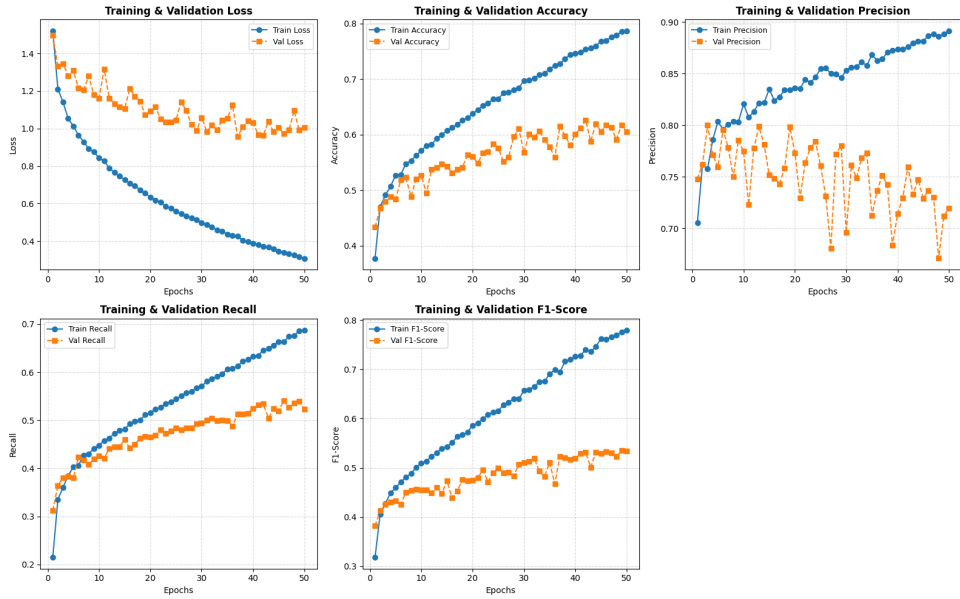


Figure 5: Training and validation metrics for the fine-tuned ResNet50 model.

The ResNet50 model achieved an overall accuracy of 0.61 on the test set. The updated per-class classification metrics are detailed in Table 2.

Table 2: Per-Class Classification Report (Fine-Tuned ResNet50)

Class	Precision	Recall	F1-Score	Support
Normal	0.80	0.63	0.71	423
Diabetic Retinopathy	0.88	0.55	0.68	369
Glaucoma	0.47	0.75	0.58	84
Cataract	0.57	0.87	0.69	31
Myopia	0.71	0.85	0.77	26
AMD	0.24	0.48	0.32	25
Hypertension	0.19	0.38	0.25	8
Others	0.24	0.54	0.33	99
Accuracy				0.61
Macro Avg	0.51	0.63	0.54	1065
Weighted Avg	0.72	0.61	0.64	1065

4 Conclusion

The fine-tuned ResNet50 model outperformed the custom CNN built from scratch, achieving an overall accuracy of 0.61 compared to 0.53. With the same number of training epochs, the ResNet50 architecture demonstrated faster and more stable convergence, which was heavily dependent on the correct application of the model-specific preprocessing function.

While the ResNet50 model successfully improved classification metrics across several minority classes (for example, raising the Hypertension F1-score to 0.25 compared to the previous run), the inherent class imbalance still affects performance. The highest precision and F1-scores remain concentrated in the heavily represented classes like Normal and Diabetic Retinopathy.

In evaluating models for real-time deployment, the custom CNN offers faster inference speeds due to its lightweight architecture, which is beneficial for constrained environments. However, the fine-tuned ResNet50 yields an 8% absolute increase in accuracy. Because diagnostic reliability is the primary concern in medical image classification, the ResNet50 model is the superior choice for deployment, provided the deployment hardware can accommodate its computational requirements.